

Sneha Hassan

snehahassan9970@gmail.com | +1 412-728-4641 | linkedin.com/in/sneha-hassan | github.com/snehassan | Pennsylvania, USA

SKILLS

Developer Education: Technical Writing, Workshop Facilitation, API Documentation, Developer Onboarding, Curriculum Design, Demo Building, Prompt Engineering

AI & LLM Tooling: Python, LangChain, PyTorch, TensorFlow, LLM Fine-tuning, RAG, Agentic Workflows, NLP, OpenCV, Claude, Cursor

Languages & APIs: TypeScript, JavaScript, Java (Spring Boot), RESTful APIs, GraphQL, Flask, Node.js, SQL

Frontend: React.js, Next.js, React Native, HTML/CSS, Redux

Cloud & DevOps: AWS, GCP, Docker, CI/CD, Git, Firebase, Linux, PostgreSQL, MongoDB, Redis

EXPERIENCE

Carnegie Mellon School of Computer Science

Aug 2025 – Dec 2025

Teaching Assistant, Product Management Essentials

Pittsburgh, PA

- Designed and delivered 2 hands-on workshops on AI-driven product discovery using LLMs and prompt engineering, guiding 20+ students from problem framing to working prototypes using Claude and GPT-4
- Designed PM curriculum covering customer discovery, JTBD frameworks, and roadmap prioritization; created reusable educational materials and documentation across multiple cohorts
- Mentored 15 students through full product lifecycles, translating complex technical and strategic concepts into actionable frameworks with 95% successful course outcomes

Carnegie Mellon University

Feb 2026 – Present

Graduate Researcher, GenAI & Bio-Imaging

Pittsburgh, PA

- Built specialized LLMs for Cryo-Electron Tomography automating cellular annotation (8 hours to 12 seconds per sample); authored architecture documentation and technical runbooks enabling research team adoption without engineering support
- Developed async evaluation framework across 296 bioinformatics benchmarks; wrote reproducible technical specifications enabling systematic comparison of fine-tuned model variants

Telvue Corporation (Broadcast Automation)

May 2025 – Aug 2025

Software Developer Intern

Mt. Laurel, NJ

- Built event-driven push notification API using Firebase Admin SDK with pub/sub triggers and retry logic; authored integration documentation that enabled the mobile team to self-serve without engineering support, increasing mobile engagement 40% across 13K+ users
- Led cross-functional API alignment across engineering, design, and mobile teams; created shared technical specifications reducing design-to-production integration time from 3 weeks to under 2 weeks

Ather Energy Ltd. (EV Scooter Company)

Oct 2023 – Nov 2024

Software Developer

Bangalore, India

- Designed, built, and documented 6 RESTful APIs in Spring Boot powering the favorite-locations feature for 100K+ users; wrote developer guides that cut location-search latency by 40% and enabled independent mobile integration
- Trained and deployed an ML pothole detection model (92% accuracy) using Python signal processing on accelerometer data; wrote technical documentation translating model behavior for non-ML engineers on the product team, driving 35% safety-feature adoption growth
- Built and maintained cross-team test automation platform in Java, raising coverage from 32% to 78% and cutting QA cycle time by 45% across 4 engineering squads

Ather Energy Ltd.

Feb 2023 – Oct 2023

Software Developer Intern

Bangalore, India

- Built full-stack campaign PWA using React and Java backend enabling real-time user participation across 10+ cities, driving 25K+ sign-ups in the first month; created onboarding documentation for partner teams integrating the campaign API

PROJECTS

Socratic LLM for Technical Pedagogy

Oct 2025 – Feb 2026

- Fine-tuned a foundation LLM with custom NLP guardrails to deliver Socratic hints over direct answers; built full-stack app (React and Flask) with under 2s response latency serving a 30-user pilot; wrote technical documentation explaining the guardrail architecture for non-ML collaborators

RecoVR – VR Rehabilitation Platform | Founder & Software Engineer, Incubated by CMU

Oct 2023 – Present

- Built Unity VR physiotherapy app with custom C# scripts and positional tracking; wrote a detailed technical README documenting architecture decisions, tradeoffs, and onboarding steps that serves as the primary resource for new contributors

Vending Analytics | Software Engineer, Sheetz

Aug 2025 – Dec 2025

- Engineered a CV pipeline (Python, OpenCV, TensorFlow) detecting empty vending slots at 94% accuracy, served via a React dashboard; demoed to Sheetz operations team and authored a technical explainer translating model outputs for non-technical stakeholders

EDUCATION

Carnegie Mellon University

Jan 2025 – Dec 2025

M.S. Engineering & Technology Innovation Management | Software and Data Focus | GPA: 3.77/4

Pittsburgh, PA

APJ Abdul Kalam Technological University

Aug 2019 – May 2023

B.Tech in Computer Science | GPA: 3.94/4

Kerala, India

LEADERSHIP & AWARDS

United Nations Millennium Fellow — Led 150 global fellows for community-impact work on SDG 8 (2022–2023)